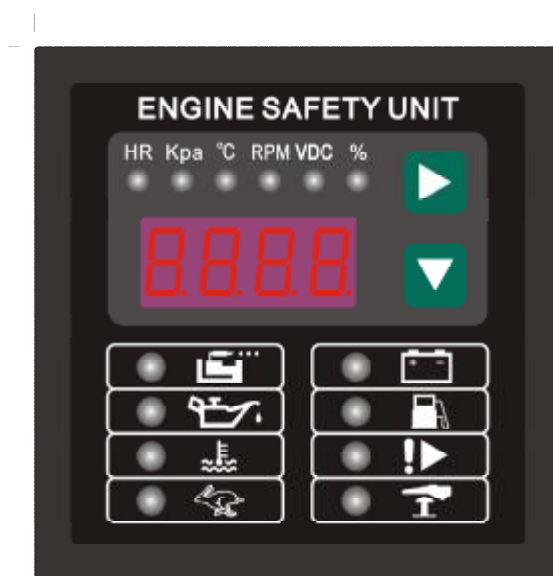


Manual

GU304 Genet Protective Device



The Interpretation of the symbol**WARNING:**

A WARNING indicates a potentially hazardous situation which, if not avoided, could result in death, serious personal injury or property damage.

**CAUTION:**

A CAUTION indicates a potentially hazardous situation which, if not avoided, could result in damage to equipment or property.

**NOTE:**

A NOTE provides other helpful information that does not fall under the warning or caution categories.

**WARNING:**

Read this entire manual pertaining to the work to be performed before installing, operating, or servicing this controller. Practice all plant and safety instructions and precautions. Failure to follow instructions can cause personal injury and/or property damage.

The engine or other type of prime mover should be equipped with an overspeed shutdown device to protect against runaway or damage to the prime mover with possible personal injury, loss of life, or property damage.

The overspeed shutdown device must be totally independent of the prime mover control system. An over temperature or low pressure shutdown device may also be needed for safety, as appropriate.

**CAUTION:**

To prevent damage to a controller that uses an alternator or battery-charging device, make sure the charging device is turned off before disconnecting the battery from the system.

**CAUTION:**

Controllers contain static-sensitive parts. Observe the following precautions to prevent damage to these parts:

Do not disassemble the rear back of controller and touch the components or conductors on a printed circuit board.

**CAUTION:**

The controller comes with the factory settings. As the factory settings may not fully meet the actual user needs, the generator settings must be checked before.

History

No.	Rev.	Date	Editor	Validation	Changes
1	HM1076ER1	2012.11.5	CHEN	P.L	NEW

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1 Description

GU304 is an engine safety module, which is used for engine safety protection, displaying running status and parameters. Module with key switch, can be used for the engine start running and stop, The module measures and displays multi-parameters for engine, if warning occurs during engine running, LED on panel will display related warning; if failure occurs, LED on panel will display related failure, module will closes fuel and engine stops. Its main characteristics are as following:

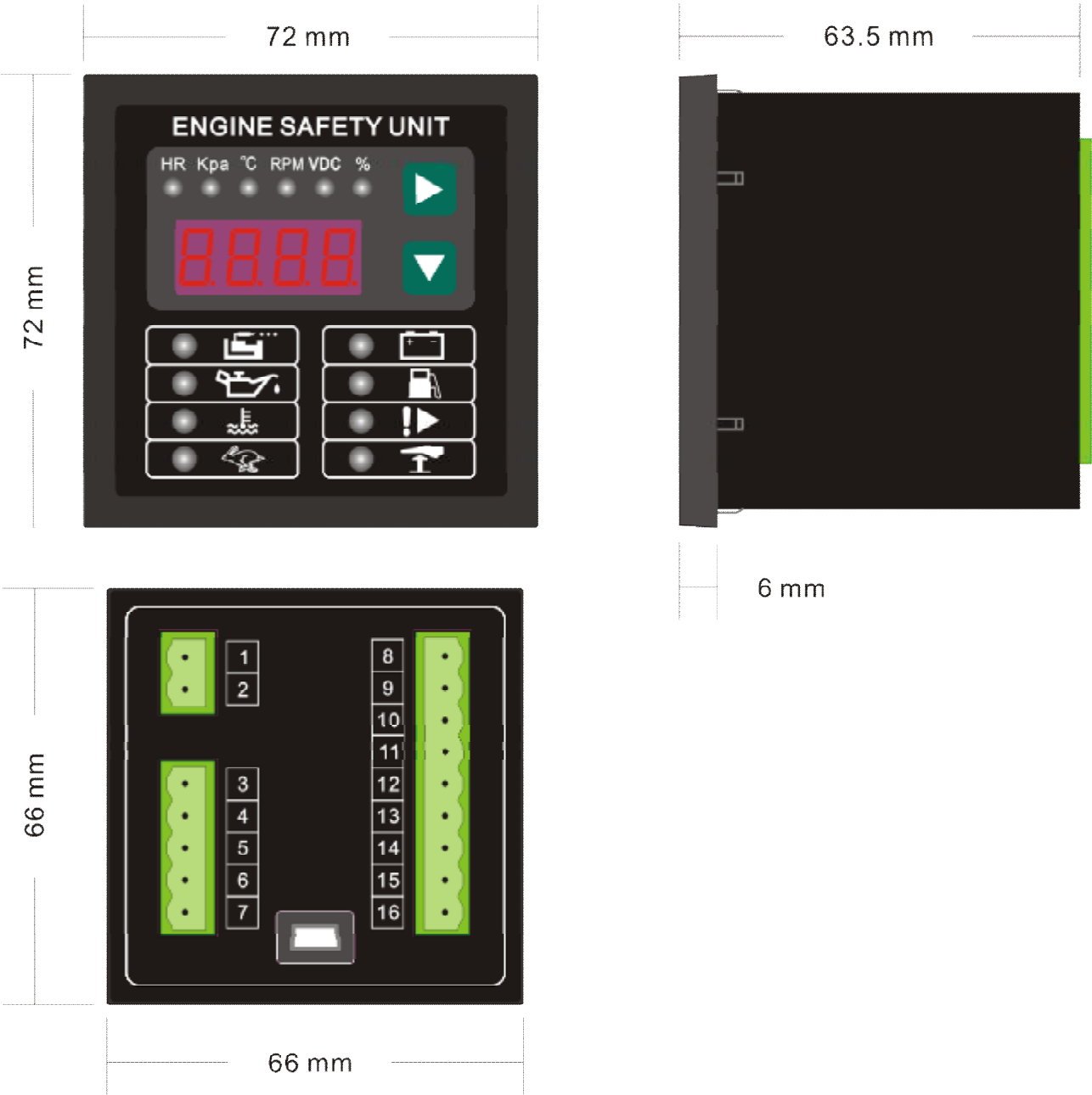
- I Adopts MPU as core.
- I The protective parameters can be configured neatly by panel buttons and LED.
- I Measure and display parameters for engine, such as battery voltage, oil pressure, coolant temperature, rpm and running hours.
- I The fuel mode of engine can be set via procedure.
- I Inset kinds of P/T sensor for user's choice.
- I Optional USB communication port to communicate with the PC, and can be read out and set the operating parameters of the controller.
- I The controller adopts DIN72 standard enclosure, all connections are connected by pin-like and locked up terminals, easier and more convenient to install, connect, move and replace the device.

This manual is only suitable for GU304 protective Device, user must carefully read this manual first.

2 The Outline Dimension Drawings and Controller Wiring

2.1 Following Details:

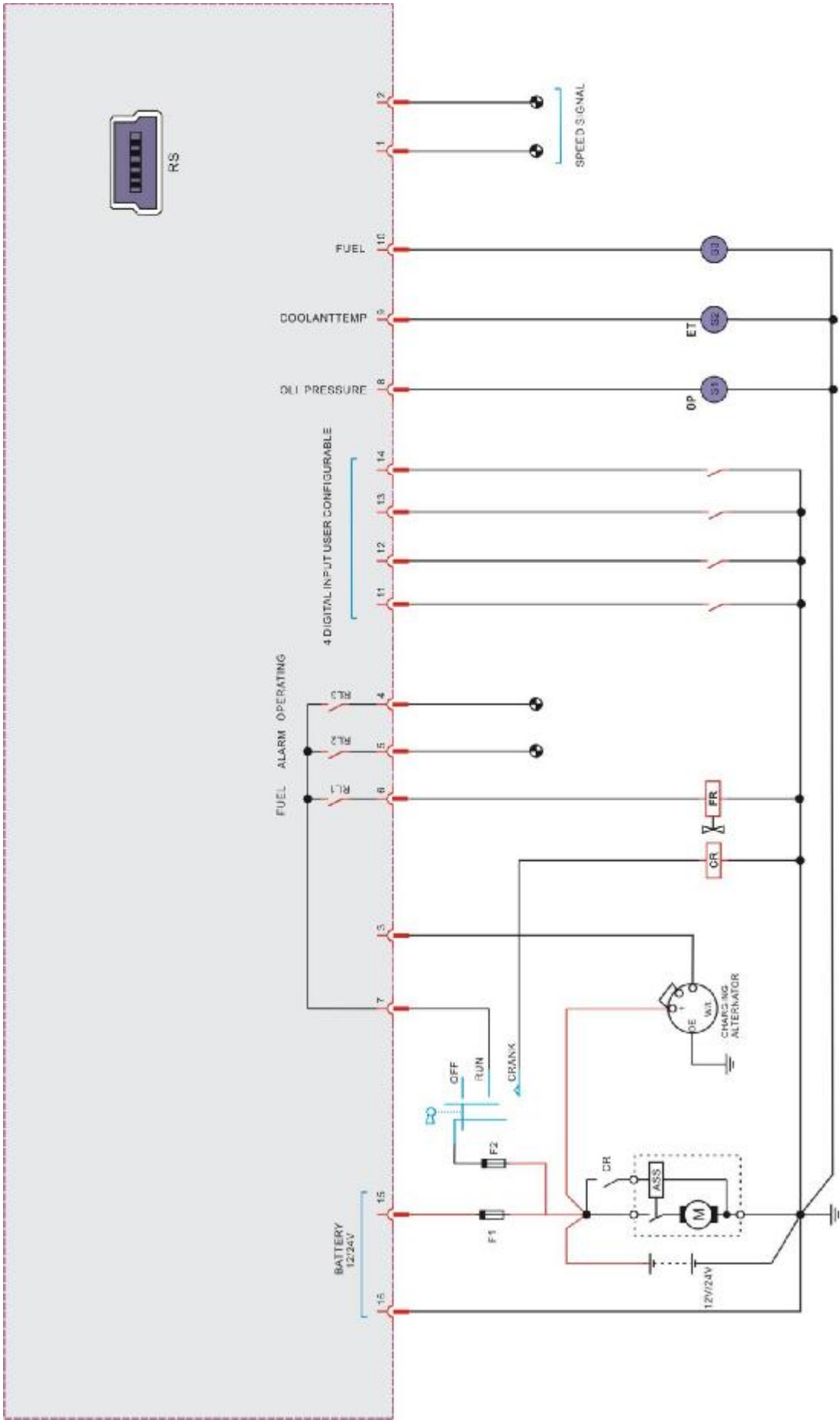
Operation Panel	W72mm×H72mm
Panel Cutout	W67mm×H67mm
Thickness	D63.5mm（without connection）



2.2 Terminal Connections:

Pin	Function Description	Signal	Dim
1	Speed signal 1	1-200Vac/1-10000Hz	Two-core shielded cable
2	Speed signal 2		
3	Charger excitation power output	if not used, do not connect to negative	1mm ²
4	Operation relay output	N.O. contact, 3A/30Vdc	1mm ²
5	Fault relay output	N.O. contact, 3A/30Vdc	1mm ²
6	Gun relay output	N.O. contact, 3A/30Vdc	1mm ²
7	Comm. port for relay		1mm ²
8	Oil pressure detect	Resistive pressure sensor (<1KΩ)	2.5mm ²
9	Temperature detect	Temperature sensor (<1KΩ)	2.5mm ²
10	Fuel level detect	Fuel level sensor (<1KΩ)	2.5mm ²
11	D-Input 1	low potential is active, Defined 1	1mm ²
12	D-Input 2	low potential is active, Defined 2	1mm ²
13	D-Input 3	low potential is active, Defined 3	1mm ²
14	D-Input 4	low potential is active, Defined 4	1mm ²
15	Battery supply (+B)	12V/24V (8-35Vdc continuous)	2.5mm ²
16	Battery supply (-B)		2.5mm ²

2.3 Typical Wiring Diagram:



3 Panel Operation

The operation panel consists of 3 sections: LCD display indicating measurement parameters, LED indicator for common failure, and push buttons for Gense.

The LCD display and its control push buttons provide a friendly operational interface for the operator to easily control the Genset, read information and parameter setting.

Control buttons and LEDs

Function Description	Tag
Scroll Button / "+" Value Increase Scroll menu for parameters display. When in parameters setting mode, this button is used to increase value.	
Parameters setting Button enter into or exit parameters setting by pressing and holding this button for 2sec.	
"-" Value Decrease When in parameters setting mode, this button is used to decrease value.	
GEN.running LED LED lights up, when the engine is running.	
Low Oil Pressure LED When a low oil pressure warning, LED flash; When a low oil pressure outage, LED illuminate.	
High Temperature LED When high temperature warning occurs, LED flash; When high temperature outage, LED illuminate.	
Overspeed LED The overspeed warning, LED flash; When speeding outage, LED illuminate.	
Charge failure LED When charging failure warning occurs, LED illuminate.	
Low Fuel level LED When a low oil level warning, LED flash; When low oil level outage, LED illuminate.	
AUX. LED Used to defined warning D-input effective, LED flash; Used to defined Outage D-input effective, LED illuminate.	
E-stop LED Used to defined E-stop switch D-input effective, LED illuminate.	

4 Measure and Display Data

Engine speed **RPM** (signal derived from magnetic pick-up or generator Hz or charger Hz)

Engine oil pressure Kpa (signal from engine LOP sensor)

Engine coolant temperature °C (signal from engine HET sensor)

Fuel Level Sensors % (signal from Fuel Level in the tank of sensor)

Battery voltage Vdc

Genset Running hour Hour

5 Parameters Setting

NO	Items		Setting Range	Preset
00	Quit			
01	Auto scroll time		1 to 60s /0 not used	5 秒
02	MPU input		0 to 1	1
03	Fly wheel teeth		1 to 999	120
04	Pair of poles		1 to 4	2
05	Fuel mode		0 Close/1 Open	0
06	Idle time		1 to 9999s /0 not used	not used
07	Safety-on delay		0 to 600s	10s
08	ALT.low preALM		1.0 to 40.0V /0.0 not used	8.0V
09	SP Over preALM		1 to 9999 RPM	1600RPM
10	SP Over alarm		1 to 9999 RPM	1710RPM
11	P-sensor type		1 to 15 /0 not used	4
12	Oil-P low preALM		1 to 1000 Kpa /0 not used	140 Kpa
13	Oil-P low alarm		1 to 1000 Kpa /0 not used	110 Kpa
14	Temperature sensor type		1 to 15 /0 not used	3
15	Coolant preALM		0 to 320℃ /9999 not used	92℃
16	Coolant alarm		0 to 320℃ /9999 not used	100℃
17	Fuel sensor type		1 to 2 /0 not used	1
18	Fuel low preALM		0 to 100% /0 not used	70%
19	Fuel low alarm		0 to 100% /0 not used	20%
20	Coolant Stop		0(warning)/1(shutdown alarm)	1
21	Critical mode		0 N/1 Y	0
22	D-Input 1 function		0 to 20	5
23	D-Input 1 logic		0 Close/1 Open	0
24	D-Input 1 delay		0 to 999s	1s
25	D-Input 2 function		0 to 20	3
26	D-Input 2 logic		0 Close/1 Open	0
27	D-Input 2 delay		0 to 999s	1s
28	D-Input 3 function		0 to 20	2
29	D-Input 3 logic		0 Close/1 Open	0
30	D-Input 3 delay		0 to 999s	1s
31	D-Input 4 function		0 to 20	4
32	D-Input 4 logic		0 Close/1 Open	0
33	D-Input 4 delay		0 to 999s	1s

Menu descriptions:**Auto scroll time**

- | LCD screen is used to set the interval of the display page, any button for 30 seconds after the start page automatically.
- | When the parameter is set to " not use" Press "▶" Manually flip.

MPU input

- | Used to define the controller for the speed measurement of input signal source.
- | When the parameter is set to "1", the controller uses the speed sensor as the engine speed measurement signal source; When the parameter is set to "0", the controller of the engine speed measurement value from the frequency of the generator signal, the conversion was calculated.
- | Speed (RPM) and frequency conversion formula: $\text{speed (RPM)} = (\text{Hz} * 60) / \text{number of pole pairs}$.
Example: the measurement of the generator frequency is 50Hz, the number of pole pairs is set to 2:00, the rotational speed (RPM) = $(50 * 60) / 2 = 1500$ (RPM).

Fly wheel teeth

- | pulses per revolution used to define engine / flywheel teeth.

Pair of poles

- | used to define the generator exciter poles.
- | controller speed measurement value from the frequency of the generator signal, for the operation of the speed measurement.

Fuel mode

- | Used to define or the definition of the engine fuel supply valve type.
- | **Start control sequence for N.C. typ:** When controller 7 # port connected to the power , controller throttle relay RL1 closed, fuel valve to open. When controller # 7 port or losing electricity or outage, controller throttle relay RL1 closed, fuel valve to open, genset stop.
- | **Start control sequence for N.O. typ:** When controller 7 # port connected to the power , controller throttle relay RL1 closed, fuel valve to open. hen controller # 7 port or losing electricity or outage, controller throttle relay RL1 closed, fuel valve to open, genset stop, the throttle relay RL1 disconnect when inside built-in failure outage time delay.

Idle time

- | The duration of engine idle running.
- | Is defined within the idle time of idling output relay closed output end of the counting, the relay restore the disconnected state.
- | When parameter is configured as "not used", idle function is inactive.

Safety-on delay

- | Used to configure the period from engine started successfully to Genset stable running. When controller 7 # port connected to the power , time delay timer is started.
- | The protection of under speed, under voltage, under frequency, low oil pressure is disabled by the controller during safety-on time delay.

**CAUTION:**

- As some of the protection are disabled during safety-on delay, so the safety-on delay should be set carefully and properly, this is very important, or it may cause engine damage.

ALT.low preALM

- Used to configure the low charger voltage value. If this parameter is configured as “not used”, the function would be inactive.
- After the safety monitoring dela, When the controller of 3 # port voltage measurement value is lower than the setting value, charge failure LED flash.

SP Over preALM

- Used to configure the over speed pre-alarm value for the engine. If this parameter is configured as “not used”, the function would be inactive.
- After the safety monitoring dela, When generator speed is higher than the set value, the pre-alarm LED will illuminate.

SP Over alarm

- Used to configure the over speed alarm value for the engine. If this parameter is configured as “not used”, the function would be inactive.
- After the safety monitoring dela, when the speed is higher than the set value, the generator stop, the prealm LED will illuminate, until controller # 7 port lost electricity.

P-sensor type

- Used to configure the type of LOP sensor.
- Optional kinds of built-in LOP sensors in controller.

Code	Mode	Note
0	not used	
1	Low oil pressure switch 1	Closed (low)
2	Low oil pressure switch 2	Open (high)
3	VDO 5 bar	
4	VDO 10 bar	
5	Datcon 7 bar	
6	Murphy 7 bar	
7	configurable 1	
8	configurable 2	
9	configurable 3	
10	configurable 4	
11	Configurable	

**CAUTION:**

- Please match the right type of the sensor or configure the right curve of the senor. Otherwise it may cause engine damage.

I The parameters appendix of LOP sensor:

VDO 5 bar:

P(Bar)	0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5
P(PSI)	0	7.3	14.5	21.8	29.0	36.3	43.5	50.8	58.0	65.3	72.5
R(Ω)	11	29	47	65	82	100	117	134	151	167	184

VDO 10 bar:

P(Bar)	0.0	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0	10.0
P(PSI)	0	14.5	29.0	43.5	58.0	72.5	87.0	101.5	116.0	130.5	145.0
R(Ω)	10	31	52	71	90	106	124	140	155	170	184

Datcon 7 bar:

P(Bar)	0.0	0.7	1.4	2.1	2.8	3.4	4.1	4.8	5.5	6.2	6.9
P(PSI)	0	10.0	20.0	30.0	40.0	50.0	60.0	70.0	80.0	90.0	100.0
R(Ω)	240	200	165	135	115	95	78	63	48	35	25

Murphy 7 bar:

P(Bar)	0.0	0.7	1.4	2.1	2.8	3.4	4.1	4.8	5.5	6.2	6.9
P(PSI)	0	10.0	20.0	30.0	40.0	50.0	60.0	70.0	80.0	90.0	100.0
R(Ω)	240	205	171	143	123	103	88	74	60	47	33

Pre-set 1:

P(Bar)	0.0	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0	10.0
P(PSI)	0	14.5	29.0	43.5	58.0	72.5	87.0	101.5	116.0	130.5	145.0
R(Ω)	15	31	49	66	85	101	117	132	149	164	178

Pre-set 2:

P(Bar)	0.0	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0	10.0
P(PSI)	0	14.5	29.0	43.5	58.0	72.5	87.0	101.5	116.0	130.5	145.0
R(Ω)	30	41	65	88	110	115	145	150	172	185	190

Pre-set 3:

P(Bar)	0	1.7	3.4	5.2	6.9	8.6	10.3
P(PSI)	0	25	50	75	100	125	150
R(Ω)	21	36	52	72	84	100	120

Pre-set 4:

P(Bar)	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0
P(PSI)	14.5	29.0	43.5	58.0	72.5	87.0	101.5	116.0	130.5
R(Ω)	195	155	127	107	88	72	61	54	48



NOTE:

- I “Configurable” means user can input the data manually according to the sensor curve.
- I When configuring, please input the “resistance- measured value” from small to big one by one.

Oil-P low preALM

- I Used to define low engine oil pressure warning threshold. When the engine oil pressure is at or below this threshold, the Oil-P low LED will illuminate.

Oil-P low alarm

- I Used to define low engine oil pressure shutdown fault threshold. When the engine oil pressure is at or below this threshold, the Oil-P low LED will illuminate, generator stop, until controller # 7 port lost electricity.

T-sensor type

- I Used to configure the type of HET sensor.
- I Optional kinds of built-in HET sensors in the controller.

Code	Type	Note
0	Not used	
1	High temperature switch 1	Closed (low)
2	High temperature switch 2	Disconnect (high)
3	VDO 120℃	
4	VDO 150℃	
5	Datcon	
6	Murphy	
7	Pt100	
8	Configurable 1	
9	Configurable 2	
10	Configurable 3	
11	Configurable 4	
12	Configurable	

**CAUTION:**

- I Please match the right type of the sensor or configure the right curve of the sensor. Otherwise it may cause engine damage.

- I The parameters appendix of HEP sensor:

VDO 120℃:

T(℃)	40	50	60	70	80	90	100	110	120	130	140
T(°F)	104	122	140	158	176	194	212	230	248	266	284
R(Ω)	291	197	134	97	70	51	38	29	22	18	15

VDO 150℃:

T(℃)	50	60	70	80	90	100	110	120	130	140	150
T(°F)	122	140	158	176	194	212	230	248	266	284	302
R(Ω)	322	221	155	112	93	62	47	37	29	23	19

Datcon:

T(℃)	40	50	60	70	80	90	100	110	120	130	140
T(°F)	104	122	140	158	176	194	212	230	248	266	284
R(Ω)	900	600	400	278	200	141	104	74	50	27	4

Murphy:

T(°C)	40	50	60	70	80	90	100	110	120	130	140
T(°F)	104	122	140	158	176	194	212	230	248	266	284
R(Ω)	1029	680	460	321	227	164	120	89	74	52	40

PT100:

T(°C)	-100	-50	0	20	40	60	80	100	150	200	300
T(°F)	-148	-58	32	68	104	140	176	212	302	392	572
R(Ω)	60	81	100	108	116	123	131	139	157	176	212

Pre-set 1:

T(°C)	20	30	40	50	60	70	80	90	100	110	120
T(°F)	68	86	104	122	140	158	176	194	212	230	248
R(Ω)	900	600	420	282	152	113	86	62	48	40	30

Pre-set 2:

T(°C)	30	50	60	70	80	90	100	110	120
T(°F)	86	122	140	158	176	194	212	230	248
R(Ω)	980	400	265	180	125	90	65	50	38

Pre-set 3:

T(°C)	20	30	40	50	60	70	80	90	100	110	120
T(°F)	68	86	104	122	140	158	176	194	212	230	248
R(Ω)	805	540	380	260	175	118	83	58	42	30	21

Pre-set 4:

T(°C)	28	35	40	50	60	70	80	90	95	98
T(°F)	82	95	104	122	140	158	176	194	203	208
R(Ω)	579	404	342	250	179	136	103	77	67	63

**NOTE:**

- I “Configurable” means user can input the data manually according to the sensor curve.
- I When configuring, please input the “resistance-value” from small to big one by one.

Coolant preALM

- I Used to define engine coolant preALM threshold. When the engine oil pressure is at or below this threshold, the coolant LED will illuminate.

Coolant alarm

- I Used to define coolant Alarm shutdown fault threshold. When the engine oil pressure is at or above this threshold, the coolant LED will illuminate, generator outage, until controller # 7 port lost electricity.

Fuel sensor type

- I Used to define the type of fuel sensor

Optional kinds of built-in fuel sensors in the controller, such as the following menu:

Code	Type	Note
1	Configurable	Customers use applications through the communication port to configuration.
2		
3		

Fuel low preALM

- I Used to define fuel low warning thresholds. When the engine fuel low at or below the threshold, the fuel low LED will flash.

Fuel low alarm

Used to define fuel low outage threshold. When the engine fuel low at or below the threshold, the fuel low LED will illuminate, until controller # 7 port lost electricity.

Coolant Stop

- I Configurable coolant switch quantity in effective, generator is as a warning to remind or stop.

Critical mode

- I Becomes a warning, that the crew in the event of a shutdown fault alarm only non-stop in an emergency mode, all outage.

D-Input*function

- I used to define the switch input function
- I definable switch input menu is as follows:

code	Function	Note
0	Not used	
1	AUX Pre-alarm	Choose this function of D-Input is valid, AUX LED flash.
2	AUX Alarm	Choose this function of D-Input is valid, AUX LED flash, generator stop.
3	Oil pressure switch	Select this function switch input port external one mounted on the engine pressure switch, low engine oil pressure switch limit action to trigger protection. The controller allows the oil pressure switch as low oil pressure protection and low engine oil pressure protection at the same time as pressure sensor measurement. But the low oil pressure switch triggered only low oil pressure failure.

4	Temp. high switch	Select this function switch input port external to a temperature switch installed on the engine, and the limit action to trigger the switch engine temperature protection. The controller allows the temperature switch as a high-temperature protection and engine temperature protection at the same time as the measurement of the temperature sensor. But Tem. High switch only trigger temp.high outage fault.
5	Emergency stop	Select this function switch input port connected to an external emergency stop switch, when this input, the controller turns off all control output, the immediate shutdown of the engine.
6	Standby	
7	Low fuel switch	Select this function switch input port is connected to the fuel tank of the engine oil level switch for monitoring the fuel tank low oil level condition. Controller allows the oil level switch as low oil level protection and oil level sensor measurement as a low oil level protection exist at the same time, but the oil level switch triggered only low oil level of failure.
8	Standby	
9	Critical mode	Becomes a warning, that the crew in the event of a shutdown fault alarm only non-stop in an emergency mode, all outage.
10	Lamp test	Select this function switch input signal is active, the indicator light on the controller operation panel full brightness.

D-Input * logic

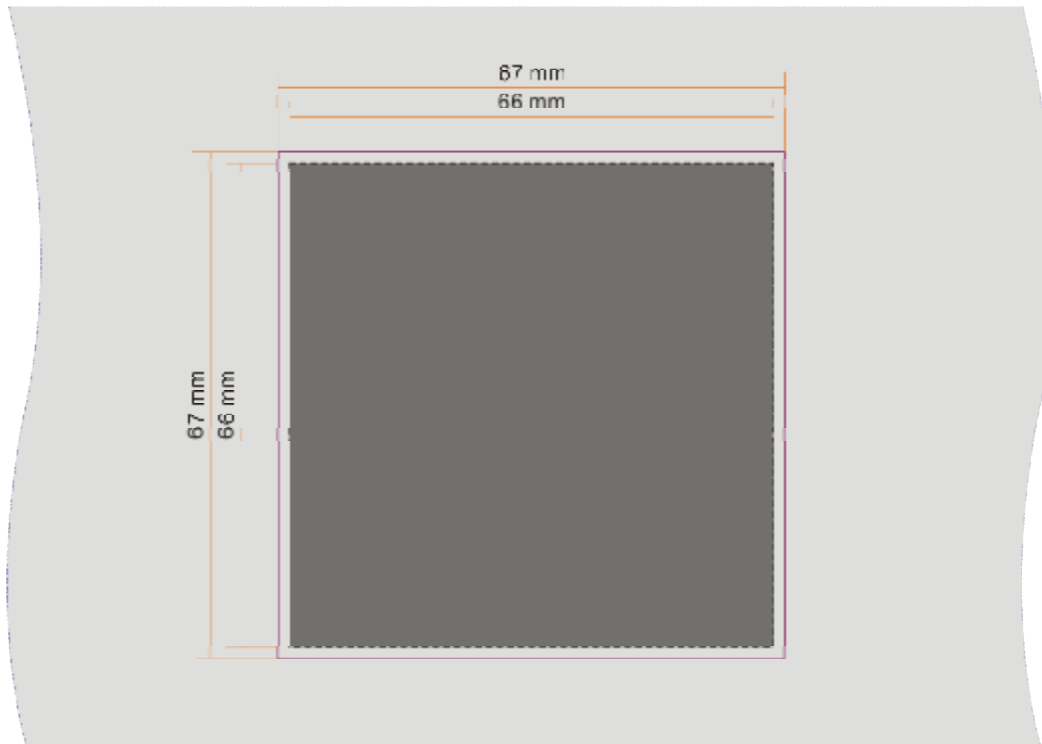
- I Used to choose the D-Input which is effectived, normally open or normally closed state.

D-Input *delay

- I If the switch quantity effective time more than the setting delay time value, trigger defined alarm level;
If the switch quantitative change as invalid before the end of the delay, delay time to zero.

6 Installation Guide

6.1 The cutout dimensional drawing installed on panel as follows:



Cutout dimension: 67mm (W) * 67mm (H). Dashed box dimensions for the controller.

The controller is fixed by 2 special fittings.



NOTE:

- ! The shock-proof equipment must be mounted if the enclosure is mounted on Genset or other heavy vibrant device.

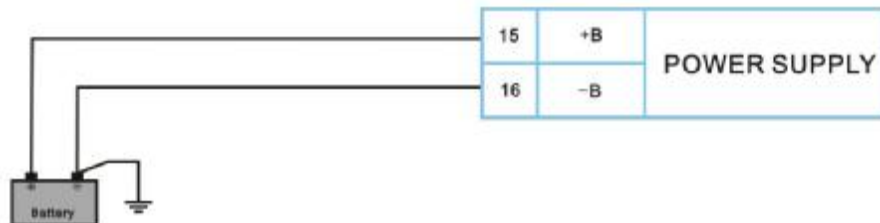
6.2 Wiring

Please refer to the above Typical Wiring Diagram for connection.

6.2.1 Power supply:

Power specification:

DC voltage range	8-35Vdc
Max. operating	@12V 120 mA, @24V 60mA
Cranking dropouts	0V for 80ms, assuming dc supply was at least 10V before dropout and recovers to 5V, controller can be normally operated dispenses with additional aux. power.



NOTE:

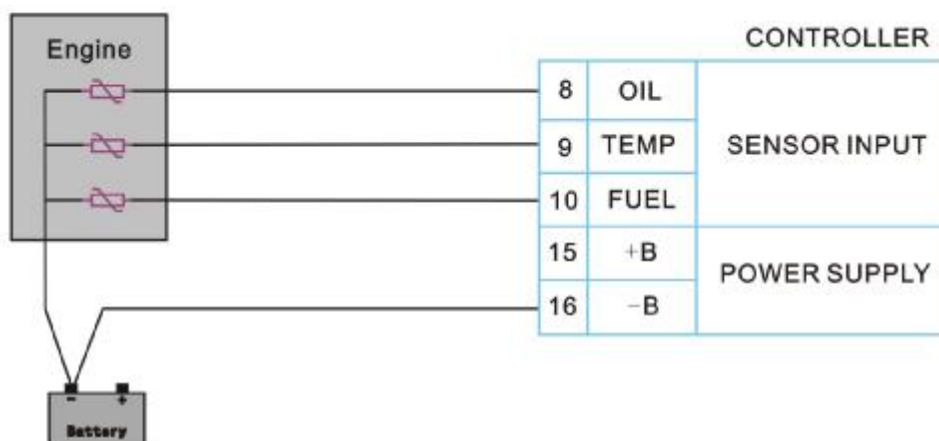
- A switch or fuse for overcurrent protection between power supply and controller must be mounted, the recommended capacity is 9A.



WARNING:

- Make sure the connection between terminal negative of controller and protective earth is good. A good ground is very important for operation of the controller, otherwise it will impact the electrical measurement and even damage the controller.

6.2.2 The installations of LOP sensor, HET sensor, and Auxiliary sensor:



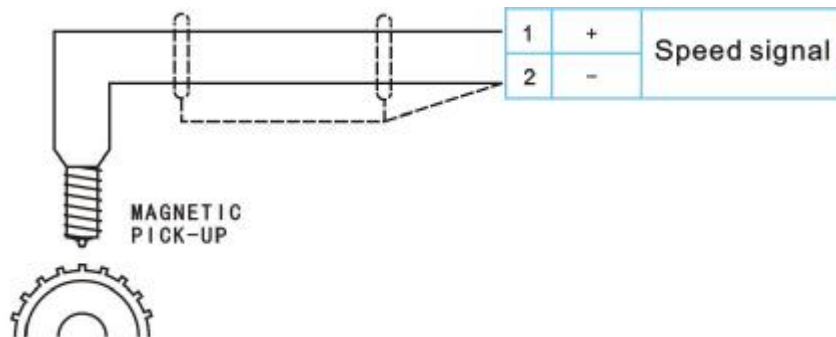
NOTE:

- Between battery to controller of the cathode cable must be special line. For ensuring the accuracy of sensor please reduce the cable resistance, the cross section area of cable should not be less than 2.5mm
- The sensor shell and engine must be well connected and do not use insulated material on sensor screw thread when installing single pole sensor.

6.2.3 The installation of MPU:

Controller to measure engine speed signal from the frequency signal of magnetic speed sensors or generator (charger).

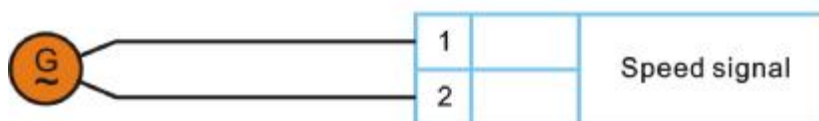
Magnetic speed sensor connection:



NOTE:

- I Shield cable must be used for connection between controller and sensor, and the shield should be earthed.

Generator or charger connection:



CAUTION:

- I Velocity measurement affects the effective value of engine crank, to avoid damage to the start motor please make sure the Ac frequency signal effectively.
- I Ac voltage cannot greater than measuring range.

7 LED displays and Menu System

7.1 Setting running parameter

Press and hold “ ” button 2sec to enter into parameter settings menu, then use “” or “” to scroll page in the same menu list, press “ ” enter into submenu, press “” or “” you can modify parameters, press “ ” to confirm modify parameters. Press and hold “ ” 2sec to quit parameter settings mode after finishing configuration

Example: (setfuel sensor type of 1)

Operation	Description
Press and hold “   ” 2sec, enter into parameters setting menu, then LCD displays:	
Press “  ” button 5 times, then LCD displays:	
Press “   ” once, Press “  ” or “  ” you can modify parameters, press “   ” to confirm modify parameters.	
press and hold “   ” 2sec will quit parameter settings menu. LED displays.	

Example: (set pair of poles of 1)

Operation	Description
Press and hold “   ” 2sec, enter into parameters setting menu, then LCD displays:	
Press “  ” button 4 times, then LCD displays:	
Press “   ” once, Press “  ” or “  ” you can modify parameters, press “   ” to confirm modify parameters.	
press and hold “   ” 2sec will quit parameter settings menu. LED displays.	

Example: (the controller is set to online programming mode)

Operation	Description
Disconnect the power supply of the controller work correctly connected via communication lines with a computer to open the programming software ABLDs.exe "computer, such as the right to open the serial port and import the upgrade process, re-pass power into the controller work program automatically update. If the operation failed power operation again.	

8 Technical Specification

8.1 Power supply:

Voltage range	12V/24V (8-35V) continuous
Max. operating current	@12V 120mA, @24V 60mA
Max. standby current	@12V 60mA, @24V 30mA
Cranking drop outs	0V for 80ms, assuming dc supply was at least 10V before dropout and recovers to 5V.
Accuracy	1%
Display	0 to 40V

8.2 Configurable digital inputs:

Number	4
Max. contact resistance	10KΩ
Max. contact current per line	0.3mA

8.3 Relay outputs

Number	3
Relay	3A/30Vdc

8.4 Charge failure input

Voltage range	0 到 35Vdc
Accuracy	1%
Max output current	@12V 120mA , @24V 240mA

8.5 Analog Inputs

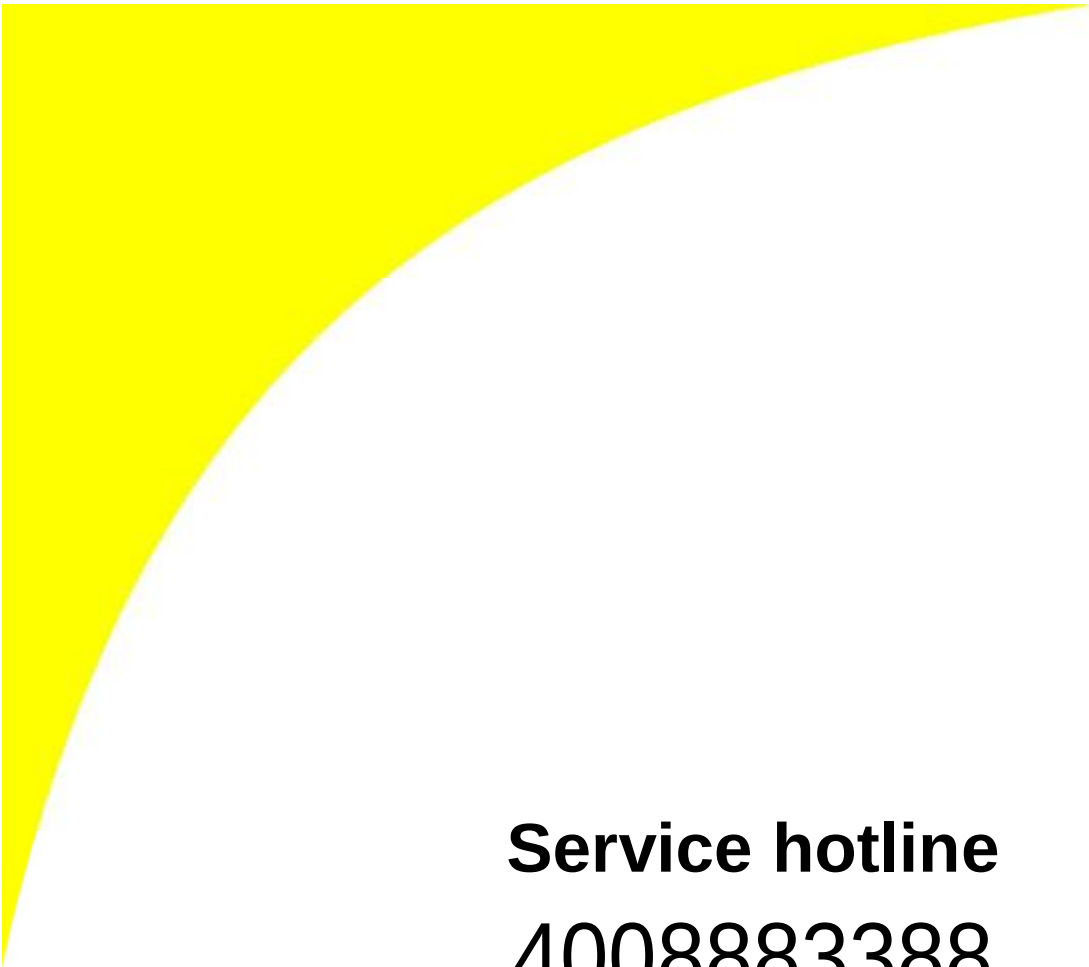
Number	3
Sensor type	Resistance
Resolution	10 bits
Range	0 to 1 KΩ
Accuracy	2% When full scale, except for sensor error

8.6 Magnetic pickup:

Voltage range	1 to 200V
Max. frequency	10000Hz
Fly wheel teeth	5 to 300

8.7 Ambient parameters

Operating ambient temperature Standards	-20 to 70°C IEC60068-2-1 and IEC60068-2-2
Storage ambient temperature Standards	-30 to 80°C IEC60068-2-1 and IEC60068-2-2
Humidity Standards	60°C, 95%RH, 48 hours IEC60068-2-30
Electro Magnetic compatibility (EMC) Standards	EN 61000-6-4 and EN 61000-6-2
Vibration Standards	EN 60068-2-6
Shock Standards	EN 60068-2-27
Electrical safety Standards	EN 60950-1
Degrees of protection Standards	IP21 (front) IP20 (back) BS EN 60529



Service hotline
4008883388

More technical support,
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